

The Physiological and Therapeutic Role of CD47 in Macrophage Function and Cancer.

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Immunotherapy is an emerging strategy in cancer therapeutics aimed at modulating the immune system to inhibit pro-tumor pathways and increase a tumor's sensitivity to chemotherapy. Several clinically approved immunotherapy treatments, such as monoclonal antibody treatments, have been successful in solid tumors such as breast, colorectal, and pancreatic. However, an outstanding challenge of these strategies is tumor cell resistance. One target of interest for immune cell modulation is targeting macrophages that enter the tumor microenvironment. More specifically, an immune checkpoint of interest is CD47. CD47 is a transmembrane protein that inhibits phagocytic activity by acting as a "don't eat me" signal. In both mice and humans, healthy cells can express CD47, while solid malignancies like colorectal and breast cancer express it most strongly. Analysis of literature data on the physiological and functional roles of tissue-resident macrophages, along with the structure and mechanisms of action of the CD47 pathway was explored. We also explored how CD47 can influence different aspects of the tumor microenvironment (i.e. cellular metabolism and hypoxia) in addition to current clinical strategies and challenges associated with targeting CD47. Overall, it was discovered that CD47 is overexpressed in a variety of cancer types in addition to normal tissue, making it a promising treatment regimen to enhance the capability of macrophages to phagocytose tumor cells. However, treatment efficacy is varied in pre-clinical and clinical models due to various challenges such as off-target effects. This review emphasizes the diverse functionality of macrophages in normal and cancerous tissue, while also emphasizing the importance of macrophage targeting and their clinical significance.

CD47 functions as a molecular switch for erythrocyte phagocytosis.

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CD47 on erythrocytes inhibits phagocytosis through interaction with the inhibitory immunoreceptor SIRP α expressed by macrophages. Thus, the CD47-SIRP α interaction constitutes a negative signal for erythrocyte phagocytosis. However, we report here that CD47 does not only function as a "do not eat me" signal for uptake but can also act as an "eat me" signal. In particular, a subset of old erythrocytes present in whole blood was shown to bind and to be phagocytosed via CD47-SIRP α interactions. Furthermore, we provide evidence that experimental aging of erythrocytes induces a conformational change in CD47 that switches the molecule from an inhibitory signal into an activating one. Preincubation of experimentally aged erythrocytes with human serum before the binding assay was required for this activation. We also demonstrate that aged erythrocytes have the capacity to bind the CD47-binding partner thrombospondin-1 (TSP-1) and that treatment of aged erythrocytes with a TSP-1-derived peptide enabled their phagocytosis by human red pulp macrophages. Finally, CD47 on erythrocytes that had been stored for prolonged time was shown to undergo a conformational change and bind TSP-1. These findings reveal a more complex role for CD47-SIRP α interactions in erythrocyte phagocytosis, with CD47 acting as a molecular switch for controlling erythrocyte phagocytosis.

Role of CD47 and Signal Regulatory Protein Alpha (SIRP α) in Regulating the Clearance of Viable or Aged Blood Cells.

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The ubiquitously expressed cell surface glycoprotein CD47 is expressed by virtually all cells in the host, where it can function to regulate integrin-mediated responses, or constitute an important part of the erythrocyte band 3/Rh multi-protein complex. In addition, CD47 can protect viable cells from being phagocytosed by macrophages or dendritic cells. The latter mechanism is dependent on the interaction between target cell CD47 and SIRP α on the phagocyte. In this context, SIRP α functions to inhibit phagocytic signaling from Fc γ receptors, complement receptors, and LDL receptor-related protein-1 (LRP-1), but not scavenger receptors. The expression level and/or distribution of CD47 may be altered on the surface of apoptotic/senescent cells, rendering the phagocytosis inhibitory function of the CD47/SIRP α interaction reduced or eliminated. Instead, the interaction between these 2 proteins may serve to enhance the binding of apoptotic/senescent target cells to the phagocyte to promote phagocytosis.

The CD47-signal regulatory protein alpha (SIRP α) interaction is a therapeutic target for human solid tumors.

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CD47, a "don't eat me" signal for phagocytic cells, is expressed on the surface of all human solid tumor cells. Analysis of patient tumor and matched adjacent normal (nontumor) tissue revealed that CD47 is overexpressed on cancer cells. CD47 mRNA expression levels correlated with a decreased probability of survival for multiple types of cancer. CD47 is a ligand for SIRP α , a protein expressed on macrophages and dendritic cells. In vitro, blockade of CD47 signaling using targeted monoclonal antibodies enabled macrophage phagocytosis of tumor cells that were otherwise protected. Administration of anti-CD47 antibodies inhibited tumor growth in orthotopic immunodeficient mouse xenotransplantation models established with patient tumor cells and increased the survival of the mice over time. Anti-CD47 antibody therapy initiated on larger tumors inhibited tumor growth and prevented or treated metastasis, but initiation of the therapy on smaller tumors was potentially curative. The safety and efficacy of targeting CD47 was further tested and validated in immune competent hosts using an orthotopic mouse breast cancer model. These results suggest all human solid tumor cells require CD47 expression to suppress phagocytic innate immune surveillance and elimination. These data, taken together with similar findings with other human neoplasms, show that CD47 is a commonly expressed molecule on all cancers, its function to block phagocytosis is known, and blockade of its function leads to tumor cell phagocytosis and elimination. CD47 is therefore a validated target for cancer therapies.

The interaction between signal regulatory protein alpha (SIRP α) and CD47: structure, function, and therapeutic target.

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CD47 is a broadly expressed membrane protein that interacts with the myeloid inhibitory immunoreceptor SIRP α (also termed CD172a or SHPS-1). SIRP α is the prototypic member of the SIRP paired receptor family of closely related SIRP proteins. Engagement of SIRP α by CD47 provides a downregulatory signal that inhibits host cell phagocytosis, and CD47 therefore functions as a "don't-eat-me" signal. Here, we discuss recent structural analysis of CD47-SIRP α interactions and implications of this for the function and evolution of SIRP α and paired receptors in general. Furthermore,

we review the proposed roles of CD47-SIRP α interactions in phagocytosis, (auto)immunity, and host defense, as well as its potential significance as a therapeutic target in cancer and inflammation and for improving graft survival in xenotransplantation.

Blocking the CD47-SIRP α axis by delivery of anti-CD47 antibody induces antitumor effects in glioma and glioma stem cells.

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Tumor initiating cells or cancer stem cells (CSCs) play an important role in the initiation, development, metastasis, and recurrence of tumors. However, traditional therapies have limited effects against CSCs and targeting these cells is crucial when developing new therapeutic strategies against cancer. One potentially targetable factor is CD47, a member of the immunoglobulin superfamily. This protein acts as an anti-phagocytic "don't eat me" signal and is often found expressed by cancer cells, particularly CSCs. CD47 functions by activating signal regulatory protein- α (SIRP- α) expressed on macrophages, preventing phagocytosis. However, the role of CD47 in glioma stem cells (GSCs) has been not been thoroughly investigated. Our study therefore examined the expression and function of this protein in glioma cells and GSCs. We found that CD47 was highly expressed on glioma cells, especially GSCs, and that expression associated with worse clinical outcomes. We also found that CD47+ glioma cells possessed stem/progenitor cell-like characteristics and knocking down CD47 expression resulted in a reduction in these characteristics. Treatment with anti-CD47 antibody led to increased phagocytosis of glioma cells and GSCs by macrophages. We next examined the effects of anti-CD47 antibody on glioma cells/GSCs in an immune competent mouse glioma model, revealing significant inhibition of tumor growth and prolonged survival times. Importantly, there were no apparent side effects in the animal model. In summary, we have shown that CD47 is a potentially safe and effective therapeutic target for glioma.

Targeting CD47 in Anaplastic Thyroid Carcinoma Enhances Tumor Phagocytosis by Macrophages and Is a Promising Therapeutic Strategy.

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Background: Anaplastic thyroid carcinoma (ATC) is one of the most aggressive human cancers, with a median survival of only three to six months. Standard treatment options and even targeted therapies have so far failed to improve long-term overall survival. Thus, novel treatment modalities for ATC, such as immunotherapy, are urgently needed. CD47 is a "don't eat me" signal, which prevents cancer cells from phagocytosis by binding to signal regulatory protein alpha on macrophages. So far, the role of macrophages and the CD47-signal regulatory protein alpha signaling axis in ATC is not well understood. Methods: This study analyzed 19 primary human ATCs for macrophage markers, CD47 expression, and immune checkpoints by immunohistochemistry. ATC cell lines and a fresh ATC sample were assessed by flow cytometry for CD47 expression and macrophage infiltration, respectively. CD47 was blocked in phagocytosis assays of co-cultured macrophages and ATC cell lines. Anti-CD47 antibody treatment was administered to ATC cell line xenotransplanted immunocompromised mice, as well as to tamoxifen-induced ATC double-transgenic mice. Results: Human ATC samples were heavily infiltrated by CD68- and CD163-expressing tumor-associated macrophages (TAMs), and expressed CD47 and calreticulin, the dominant pro-phagocytic molecule. In addition, ATC tissues expressed the immune

checkpoint molecules programmed cell death 1 and programmed death ligand 1. Blocking CD47 promoted the phagocytosis of ATC cell lines by macrophages in vitro. Anti-CD47 antibody treatment of ATC xenotransplanted mice increased the frequency of TAMs, enhanced the expression of macrophage activation markers, augmented tumor cell phagocytosis, and suppressed tumor growth. In double-transgenic ATC mice, CD47 was expressed on tumor cells, and blocking CD47 increased TAM frequencies. Conclusions: Targeting CD47 or CD47 in combination with programmed cell death 1 may potentially improve the outcomes of ATC patients and may represent a valuable addition to the current standard of care.

Exosome-SIRP α , a CD47 blockade increases cancer cell phagocytosis.

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CD47, a "don't eat me" signal, is over-expressed on the surface of most tumors that interacts with signal regulatory protein α (SIRP α) on phagocytic cells. By engaging SIRP α , CD47 limits the ability of macrophages to engulf tumor cells, which acts as a major phagocytic barrier. In this study, we developed an exosome-based immune checkpoint blockade that antagonizes the interaction between CD47 and SIRP α . These exosomes harboring SIRP α variants (SIRP α -exosomes) were sufficient to induce remarkably augmented tumor phagocytosis, lead to prime effective anti-tumor T cell response. Given that clustering of native CD47 provides a high binding avidity to ligate dimerized SIRP α on macrophage, nature-derived exosomes could be appreciable platform to antagonize CD47. Disruption of CD47-SIRP α interaction by SIRP α -exosomes leads to an increase in cells being engulfed by macrophages and a concomitant inhibition of tumor growth in tumor-bearing mice. Moreover, SIRP α -exosomes therapy promotes an intensive T cell infiltration in syngeneic mouse models of cancer, raising the possibility of CD47-targeted therapies to unleash both an innate and adaptive anti-tumor response. Note that very small amount of exosomal SIRP α proteins could effectively lead to phagocytic elimination of tumor cells both in vitro and in vivo. Our results suggest that superlative exosome-based platform has broad potential to maximize the therapeutic efficacy of membrane-associated protein therapeutics.

CD47-SIRP α -targeted therapeutics: status and prospects.

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CD47 is a "don't eat me" signal to phagocytes that is overexpressed on many tumor cells as a potential mechanism for immune surveillance evasion. CD47 and its interaction with signal-regulating protein alpha (SIRP α) on phagocytes is therefore a promising cancer target. Therapeutic antibodies and fusion proteins that block CD47 or SIRP α have been developed and have shown activity in preclinical models of hematologic and solid tumors. Anemia is a common adverse event associated with anti-CD47 treatment, but mitigation strategies-including use of a low 'priming' dose-have substantially reduced this risk in clinical studies. While efficacy in single-agent clinical studies is lacking, findings from studies of CD47-SIRP α blockade in combination with agents that increase 'eat me' signals or with antitumor antibodies are promising. Magrolimab, an anti-CD47 antibody, is the furthest along in clinical development among agents in this class. Magrolimab combination therapy in phase Ib/II studies has been well tolerated with encouraging response rates in hematologic and solid malignancies. Similar combination therapy studies with other anti-CD47-SIRP α agents are beginning to report. Based on these early clinical successes, many trials have been initiated in hematologic and solid tumors testing

combinations of CD47-SIRP α blockade with standard therapies. The results of these studies will help determine the role of this novel approach in clinical practice and are eagerly awaited.

Therapeutic Targeting of the Macrophage Immune Checkpoint CD47 in Myeloid Malignancies.

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In recent years, immunotherapies have been clinically investigated in AML and other myeloid malignancies. While most of these are focused on stimulating the adaptive immune system (including T cell checkpoint inhibitors), several key approaches targeting the innate immune system have been identified. Macrophages are a key cell type in the innate immune response with CD47 being identified as a dominant macrophage checkpoint. CD47 is a "do not eat me" signal, overexpressed in myeloid malignancies that leads to tumor evasion of phagocytosis by macrophages. Blockade of CD47 leads to engulfment of leukemic cells and therapeutic elimination. Pre-clinical data has demonstrated robust anti-cancer activity in multiple hematologic malignancies including AML and myelodysplastic syndrome (MDS). In addition, clinical studies have been underway with CD47 targeting agents in both AML and MDS as monotherapy and in combination. This review will describe the role of CD47 in myeloid malignancies and pre-clinical data supporting CD47 targeting. In addition, initial clinical data of CD47 targeting in AML/MDS will be reviewed, and including the first-in-class anti-CD47 antibody magrolimab.
